

Lecture 6: Regression Analysis

L2 - Statistics

Gustave Kenedi

2026-03-16

Recap quiz



Or click here: [link to Wooclap](#)

Today's lecture

The big question: How can we quantify and test the relationship between two numeric variables?

1. Motivation & data
2. The Simple Linear Regression model
3. Ordinary Least Squares estimation
4. Goodness of fit (R^2)
5. Inference: standard errors & the t -test
6. Reading regression output

Motivation & data

Why regression?

So far in this course we have learned to:

- **Describe** distributions of a single variable (mean, variance, histograms)
- **Quantify uncertainty** around estimates (confidence intervals)
- **Test hypotheses** about population parameters (t-tests, z-tests)

But we haven't yet answered: *How do two variables relate to each other and how can we use one to predict the other?*

Regression is the workhorse tool for answering exactly this question. It lets us: **summarise** a relationship with a line, **estimate** the average change in y for a one-unit change in x , and **test** whether that relationship is statistically significant.

Class size and student performance

A classic policy question: Does reducing class size improve student learning?

- Smaller classes cost money (more teachers, more classrooms) but may improve outcomes.
- Class size reduction has been at the heart of education policy debates for decades.

Our empirical application: Angrist & Lavy (1999)

The dataset

Test scores and class/school characteristics for **5th graders** (10/11 years old) in Jewish public elementary schools in **Israel in 1991**.

- $n \approx 2,000$ classes
- National standardised tests in **mathematics** and **reading** (Hebrew), scaled 1–100
- Key variables: **classsize** (class size), **avgmath** (avg. math score), **avgverb** (avg. verbal score)

Your turn! #1

07:00

1. Load the data below as `grades`. (Hint: `read_dta()` from the `haven` package reads Stata `.dta` files.)

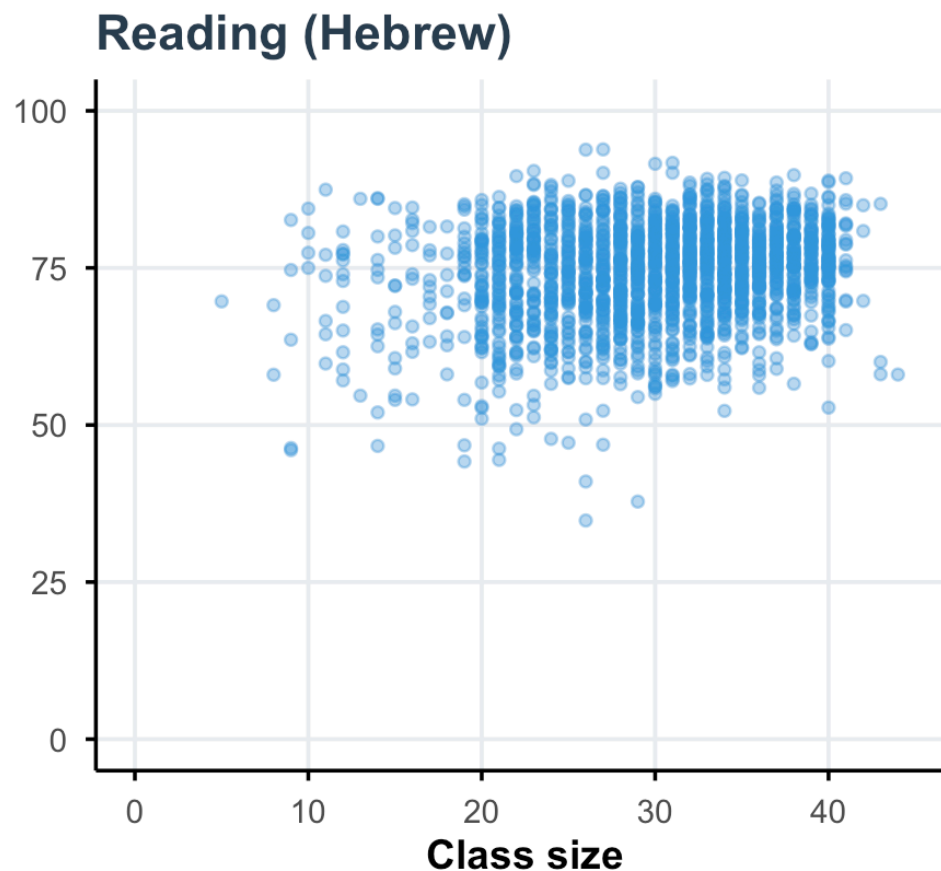
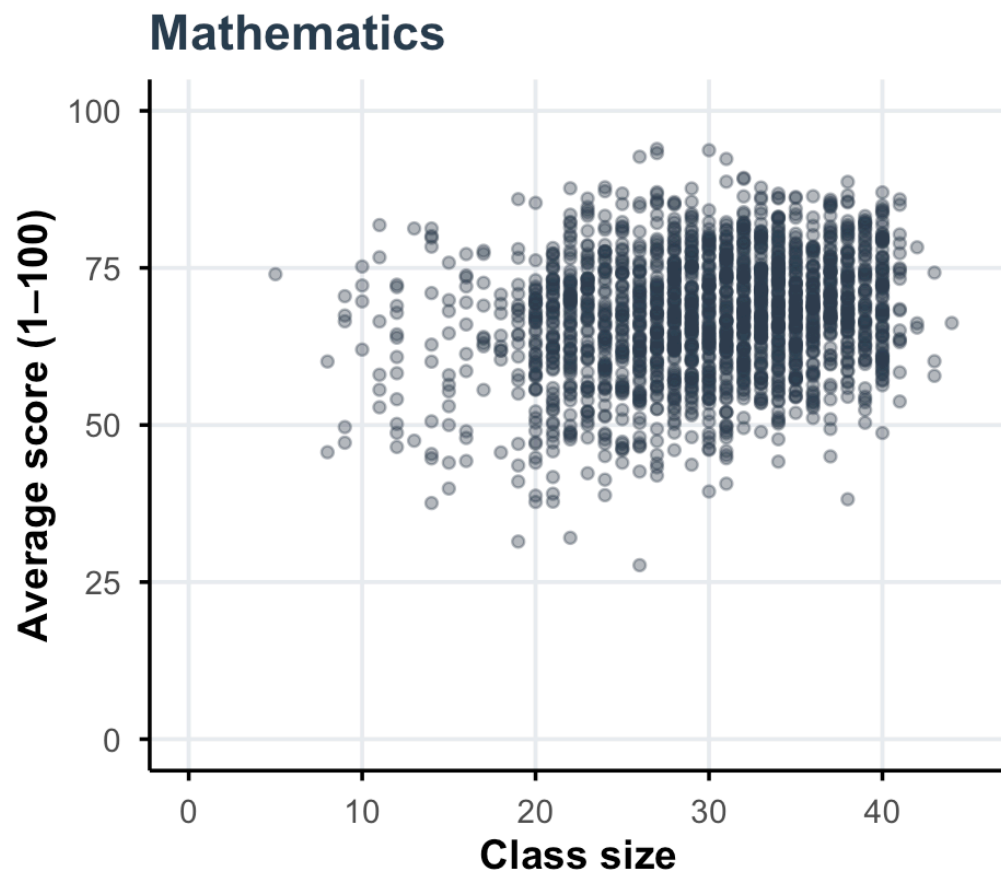
```
1 library(haven)
2 grades <- read_dta("https://www.dropbox.com/s/wwp2cs9f0dubmhr/grade5.dta?dl=1")
```

2. Describe the dataset:

- What is the unit of observation (what does each row represent)?
- How many observations are there?
- View the dataset. What variables do we have?
- Use `skim()` from the `skimr` package on `classsize`, `avgmath`, and `avgverb`. (Hint: `select()` the three variables and pipe to `skim()`.)

3. Do you have any prior intuitions about the relationship between class size and student achievement? What would you do to get a first visual insight?
4. Compute the correlation between `classsize` and both `avgmath` and `avgverb`. Is the relationship positive/negative? Strong/weak?

Class size and student performance: scatter plots



What do you notice? Is there a clear pattern?

→ For simplicity, we will **focus on mathematics** for the rest of this lecture.

Correlation

Recall from Lecture 2: **correlation** measures the *linear* association between two variables, $r \in [-1, 1]$.

```
1 cor(grades$classize, grades$avgmath)
```

```
[1] 0.2171555
```

Note

The correlation is slightly positive: larger classes are *weakly* associated with *higher* math scores. Surprising?

Correlation gives us a single number, but it doesn't answer:

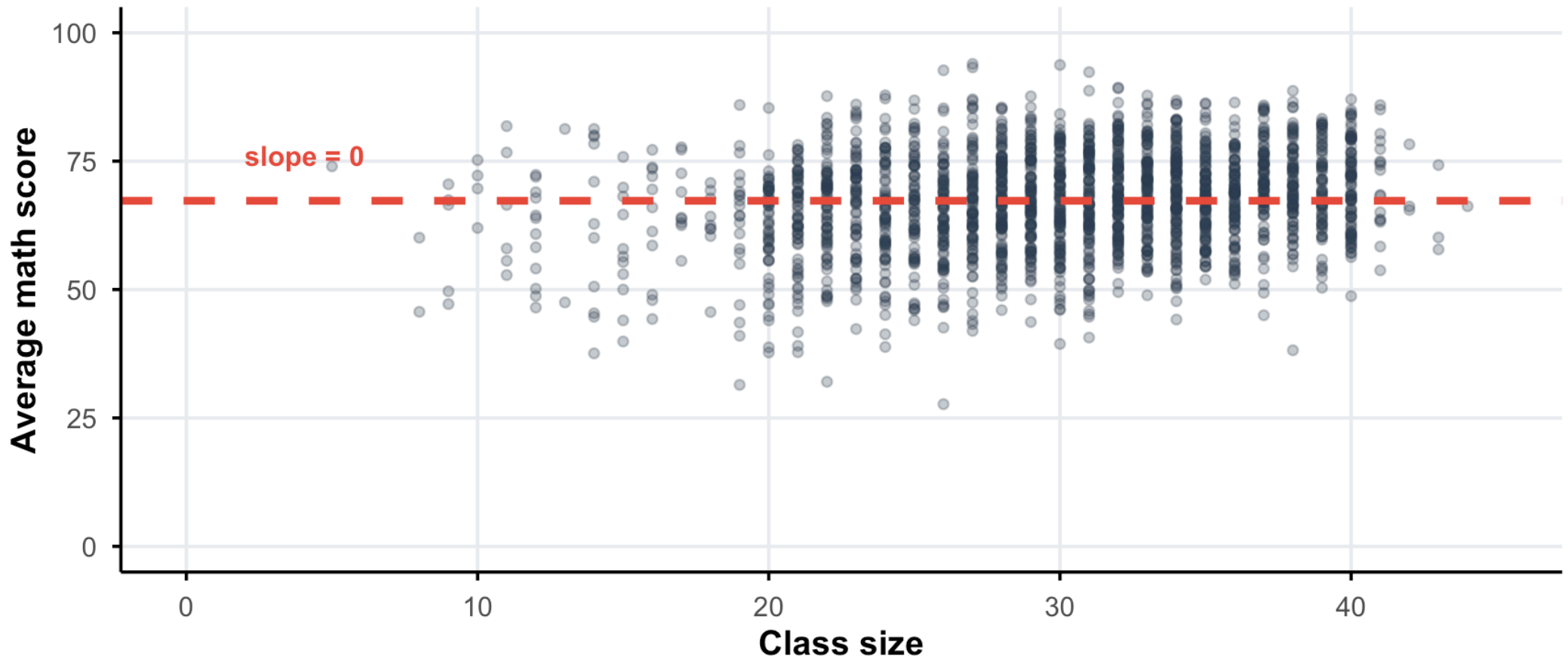
- By *how much* does the average math score change with a 1-student increase in class size?
- Is this relationship statistically different from zero?

→ We need **regression**.

Class size and math score: which line fits best?

Option 1: flat line — math scores don't change with class size

Each point is one class, Angrist & Lavy (1999)

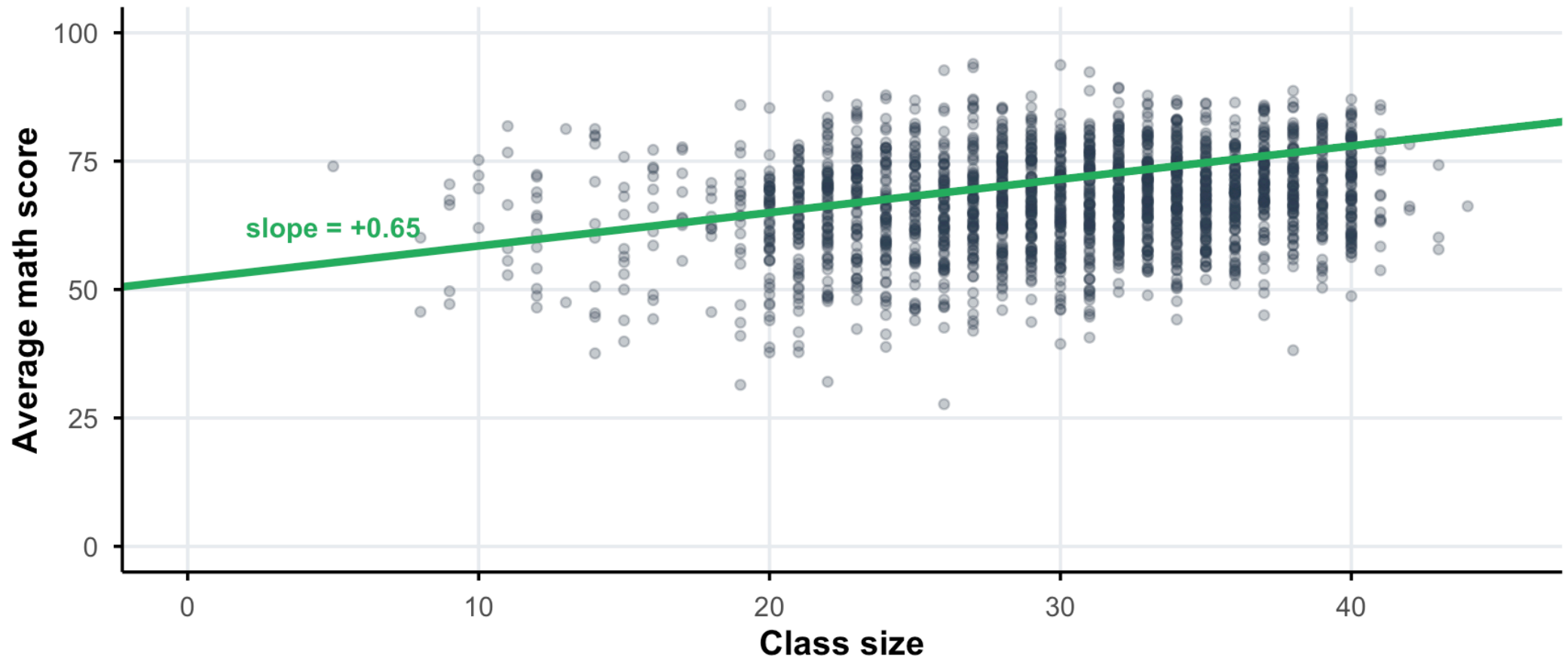


But the correlation is **positive** so we should have a positive **slope**

Class size and math score: which line fits best?

Option 2: positive slope — larger classes score higher?

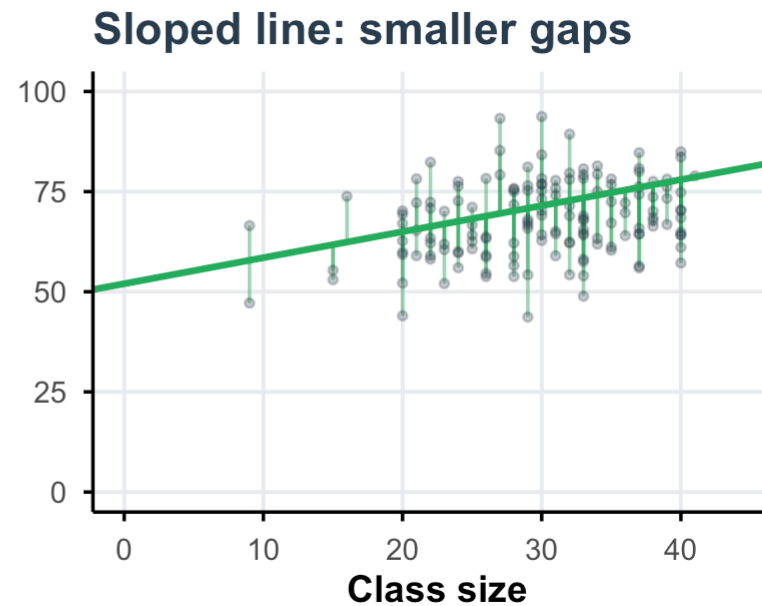
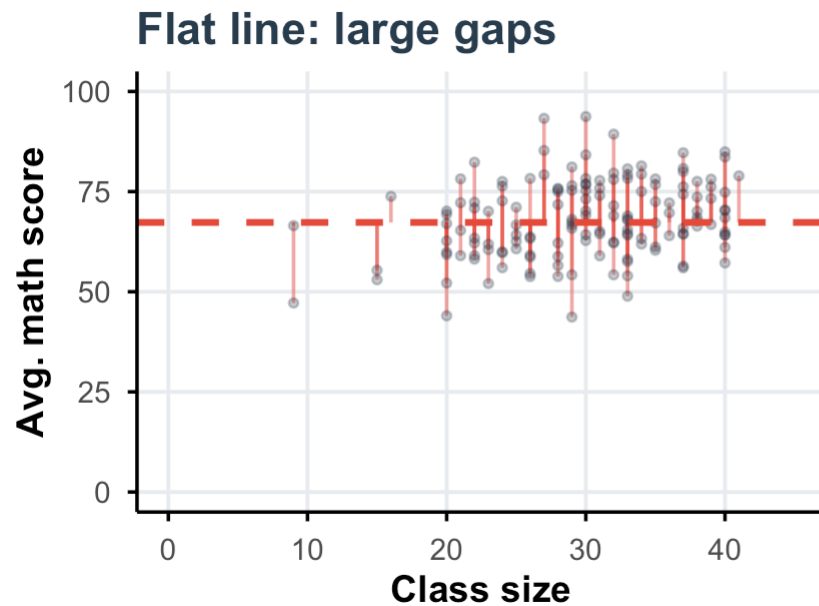
Each point is one class, Angrist & Lavy (1999)



Positive relationship, but perhaps too **steep**? → we need a **principled rule**

What makes a line “good”?

Intuitively, a good line should be **close to the data points**: the vertical gaps between each actual score and the predicted score should be small.



The sloped line *looks* better, but why exactly? And how do we find the *best* possible line? → **Ordinary Least Squares (OLS)** answers this formally.

Today's lecture

The big question: How can we quantify and test the relationship between two numeric variables?

1. Motivation & data
- 2. The Simple Linear Regression model**
3. Ordinary Least Squares estimation
4. Goodness of fit (R^2)
5. Inference: standard errors & the t -test
6. Reading regression output

Simple Linear Regression

From scatter plot to a line

We want to **summarise** the relationship between class size (x) and average math score (y) with a line.

! Two variables

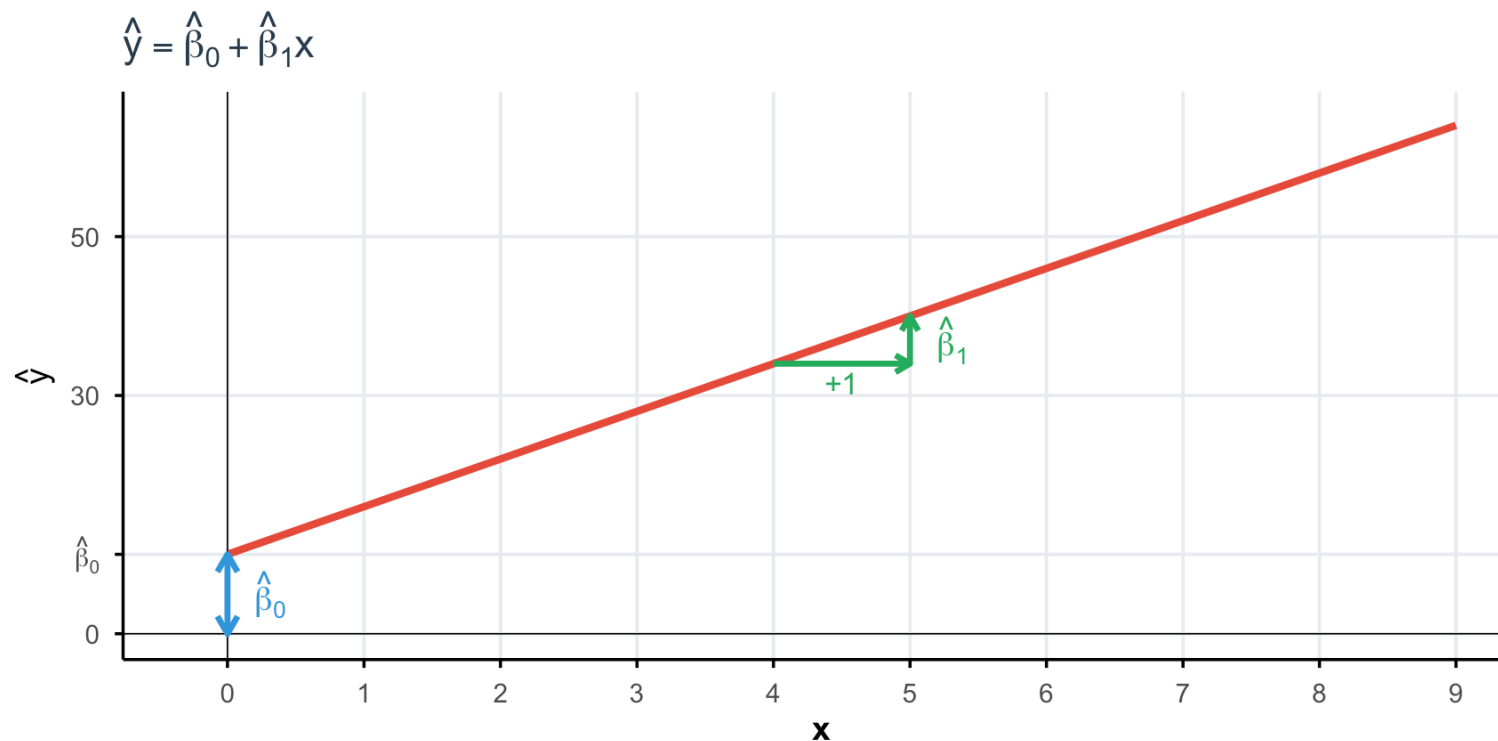
- **Outcome (dependent) variable:** $y_i \rightarrow$ average math score of class i
- **Explanatory (independent) variable:** $x_i \rightarrow$ class size of class i

A line with **intercept** $\hat{\beta}_0$ and **slope** $\hat{\beta}_1$ predicts:

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

- $\hat{y}_i$ is our **prediction** of y_i given x_i
- **Which** line? There are infinitely many $\rightarrow$ we need a rule to pick the best one.

Intercept and slope: a refresher



Definition

Intercept $\hat{\beta}_0$

Predicted value of y when $x = 0$

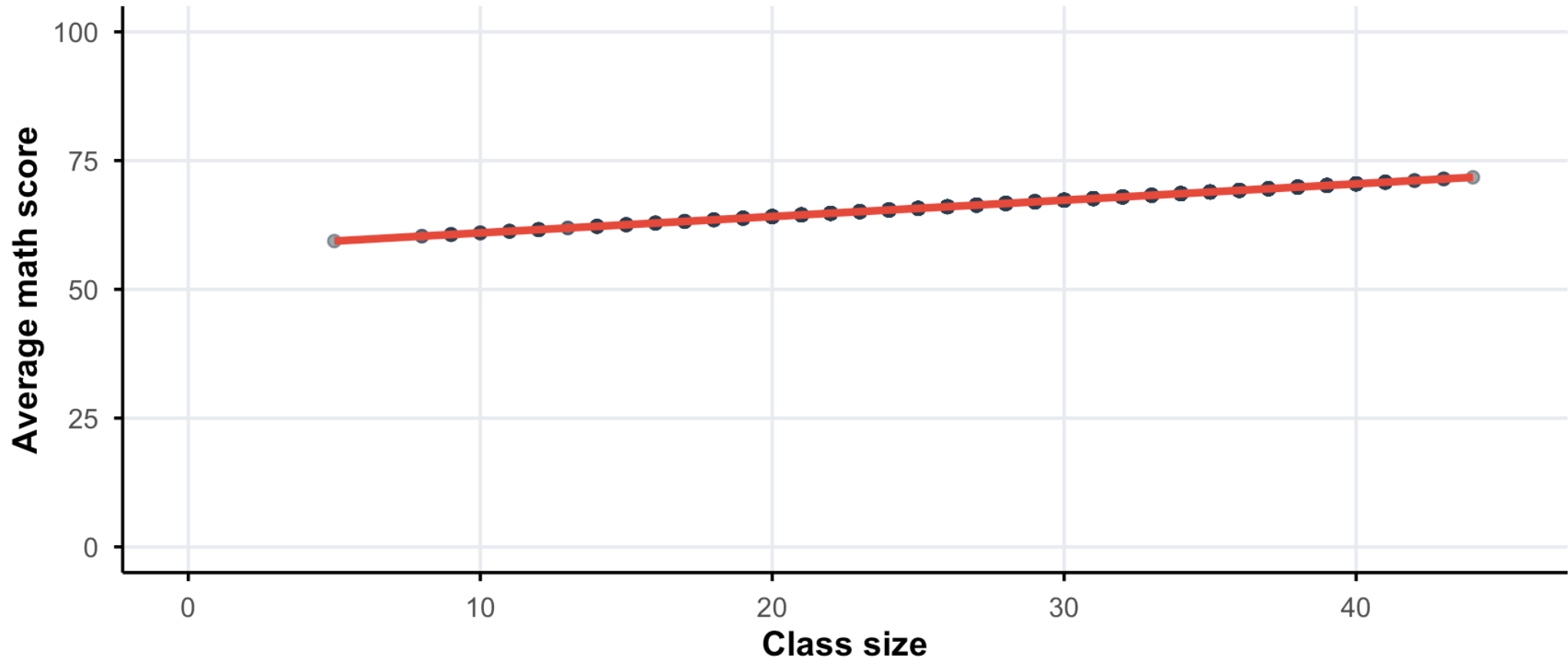
Slope $\hat{\beta}_1$

Predicted change in y for a **1-unit** increase in x

If x perfectly explained y ...

Perfect fit: every point lies exactly on the line

If class size fully determined math scores, there would be no scatter



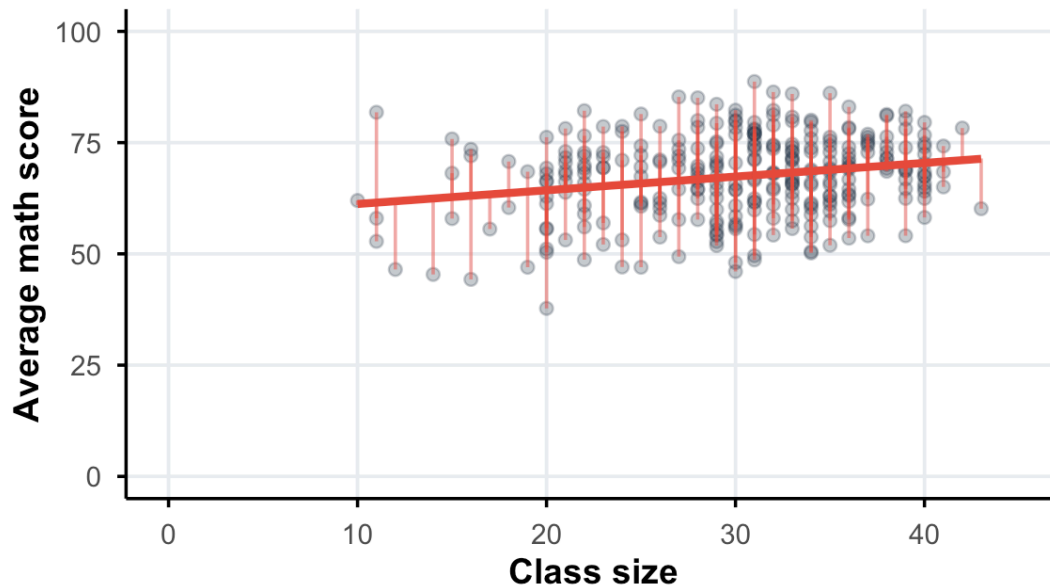
In this world: $y_i = \hat{\beta}_0 + \hat{\beta}_1 x_i \rightarrow$ **no unexplained variation.**

The residual

In practice, a student's math score depends on many things beyond class size.

Reality: points scatter around the line

Red segments = gaps between actual and predicted



We call each gap a **residual**:

$$e_i = y_i - \hat{y}_i$$

The actual value decomposes as:

$$y_i = \underbrace{\hat{\beta}_0 + \hat{\beta}_1 x_i}_{\text{prediction}} + \underbrace{e_i}_{\text{residual}}$$

i Note

The residual e_i captures everything that affects y_i **other than** x_i (talent, family background, teaching quality, etc).

Today's lecture

The big question: How can we quantify and test the relationship between two numeric variables?

1. Motivation & data
2. The Simple Linear Regression model
- 3. Ordinary Least Squares estimation**
4. Goodness of fit (R^2)
5. Inference: standard errors & the t -test
6. Reading regression output

OLS Estimation

Which line? The OLS criterion

We need a rule to choose $(\hat{\beta}_0, \hat{\beta}_1)$. A natural goal: make the residuals as **small as possible**.

Problem with summing residuals directly: positive and negative errors cancel.

Solution: Minimise the **Sum of Squared Residuals (SSR)**:

$$\text{SSR} = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

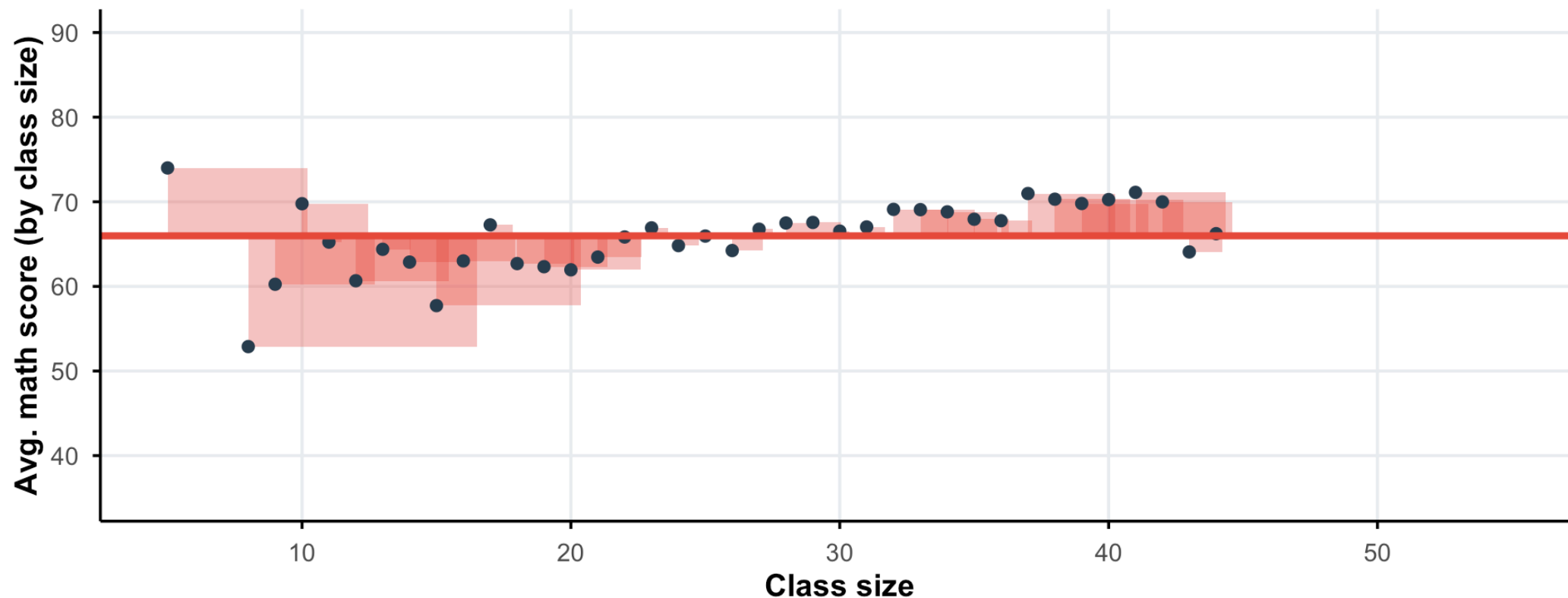
⚠ Ordinary Least Squares (OLS) Definition

OLS chooses $(\hat{\beta}_0, \hat{\beta}_1)$ to **minimise** the sum of squared residuals.

OLS: squared residuals visually

Flat line (slope = 0) — SSR = 618

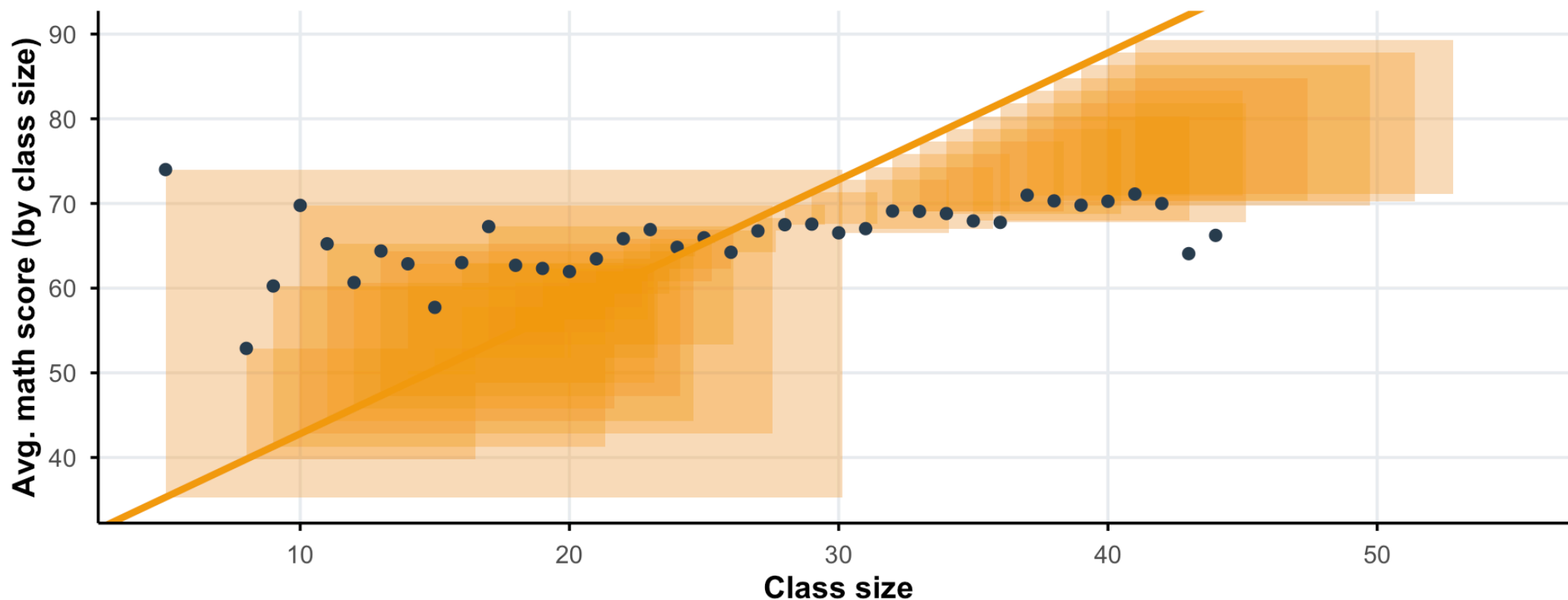
Each square's area = squared residual for that point



OLS: squared residuals visually

Too steep (slope = 1.5) — SSR = 8,408

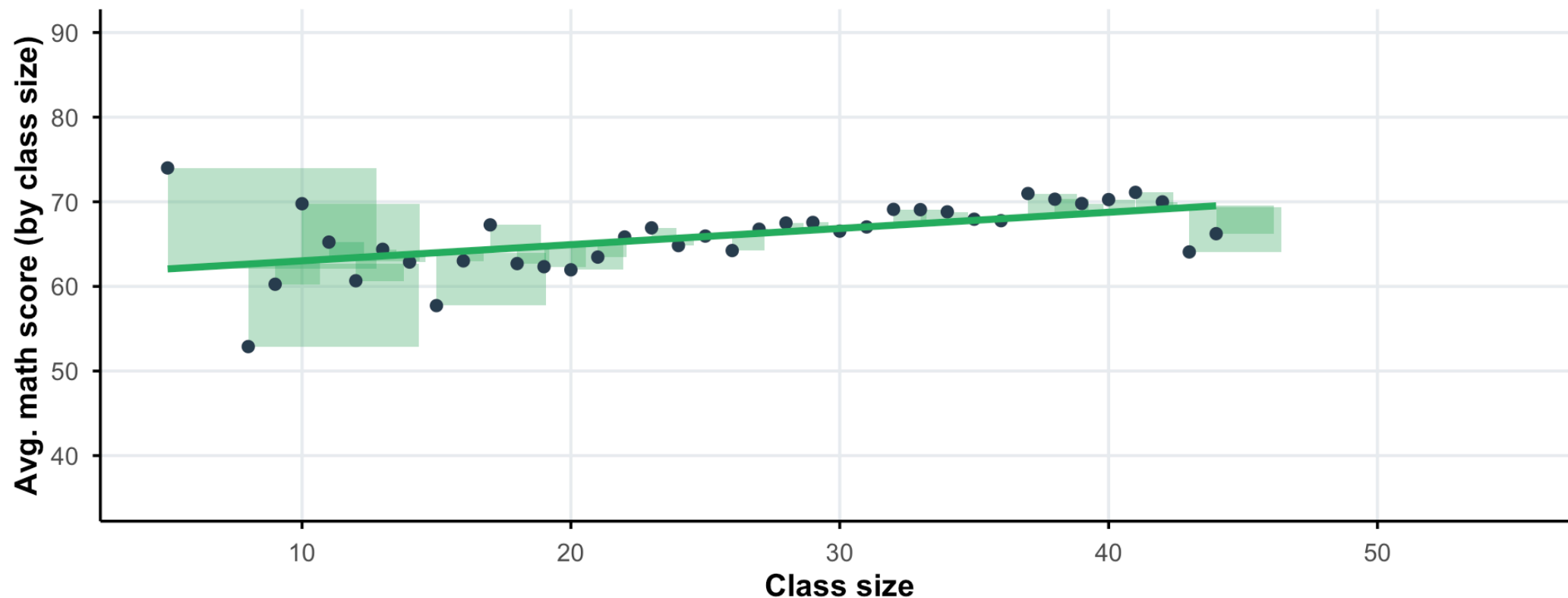
Each square's area = squared residual for that point



OLS: squared residuals visually

OLS line — SSR = 448 — minimum

Each square's area = squared residual for that point



OLS coefficient formulas

Solving the minimisation problem algebraically yields unique formulas:

! OLS formulas

$$\hat{\beta}_1 = \frac{Cov(x, y)}{Var(x)} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \cdot \bar{x}$$

Key observations:

- $\hat{\beta}_1$ is the ratio of the *covariance* of x and y to the *variance* of x
- The line always passes through the point $(\bar{x}, \bar{y})$: the means of both variables
- The formulas have unique solutions

OLS in R: `lm()`

```
1 reg <- lm(avgmath ~ classsize, data = grades)
2 reg
```

Call:

```
lm(formula = avgmath ~ classsize, data = grades)
```

Coefficients:

(Intercept)	classsize
57.7939	0.3175

So our estimated line is:

$$\widehat{\text{avg. math score}}_i = 57.79 + 0.32 \times \text{class size}_i$$

Interpreting OLS coefficients

$$\widehat{\text{avg. math score}}_i = 57.79 + 0.32 \times \text{class size}_i$$

⚠ Interpretation rules

- **Intercept** $\hat{\beta}_0$: The *predicted* average math score when class size = 0.
 - Here: $\hat{\beta}_0 = 57.79$, not economically meaningful (class size = 0 is impossible), but mathematically necessary.
- **Slope** $\hat{\beta}_1$: The *predicted* change in average math score *associated with* a **1-student increase** in class size.
 - Here: $\hat{\beta}_1 = 0.32$, a 1-student increase in class size is **associated** with a 0.32 point increase in average math score, on average.

⚠ Correlation $\neq$ Causation

We use the word **associated**, not *causes*. This positive relationship does not mean larger classes *improve* performance; there may be confounding factors.

Making predictions

$$\widehat{\text{avg. math score}}_i = 57.79 + 0.32 \times \text{class size}_i$$

Q: What is the predicted average math score for a class of **30 students**?

$$\hat{y}_{x=30} = 57.79 + 0.32 \times 30 = 67.32$$

Q: For a class of **20 students**?

$$\hat{y}_{x=20} = 57.79 + 0.32 \times 20 = 64.14$$

Q: What is the predicted *difference* between a class of 30 and a class of 20?

$$\hat{y}_{x=30} - \hat{y}_{x=20} = \hat{\beta}_1 \times (30 - 20) = 0.32 \times 10 = 3.17$$

This is just $\hat{\beta}_1$ times the difference in x !

Properties of OLS estimates

OLS residuals and predictions have useful algebraic properties:

1. Residuals sum to zero: $\sum_{i=1}^n e_i = 0$

2. Predictions average to $\bar{y}$: $\frac{1}{n} \sum_{i=1}^n \hat{y}_i = \bar{y}$

3. Residuals are uncorrelated with x : $Cov(x_i, e_i) = 0$ (by construction)

4. Residuals are uncorrelated with $\hat{y}$: $Cov(\hat{y}_i, e_i) = 0$

Note

These properties hold **by construction** for any OLS line: they are not assumptions about the world, just algebraic facts that follow from the minimisation.

Your turn! #2

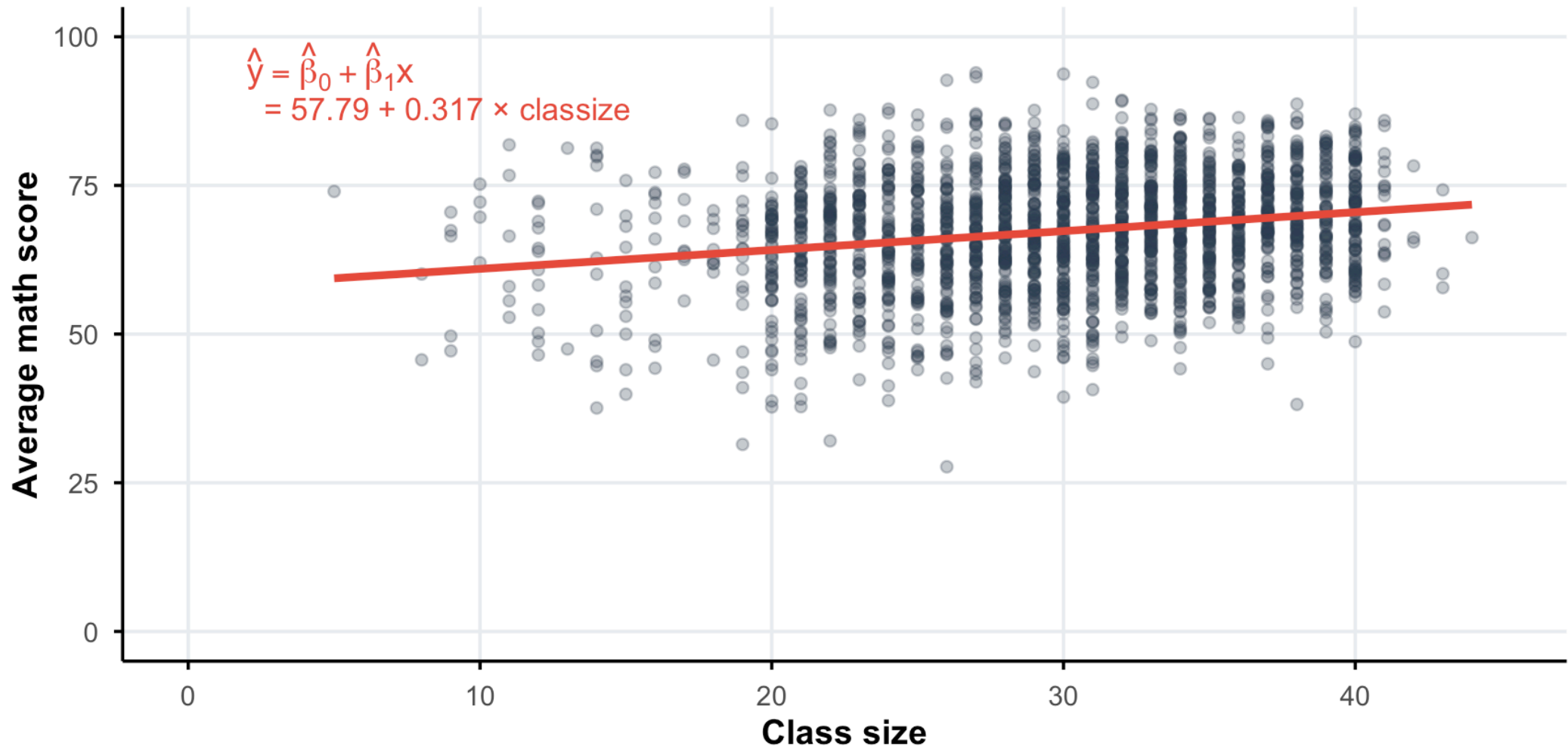
10:00

1. Regress `avgmath` on `classsize` using `lm()`. Store the result as `reg_math`. Interpret both coefficients.
2. Compute $\hat{\beta}_1$ manually using the formula $\hat{\beta}_1 = Cov(x, y) / Var(x)$. (Hint: use `cov()` and `var()`.) Then compute $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$. Do they match?
3. What is the predicted average math score for a class of 25 students? Of 35 students?
4. Repeat steps 1 and 3 for `avgverb` (verbal scores). Store the result as `reg_verb`. How do the results compare?
5. Add the regression line to your scatter plot from Your Turn #1. (Hint: use `geom_smooth(method = "lm", se = FALSE)`.)

Class size and math score: OLS fit

OLS regression line: class size and math performance

Each point is one class, Angrist & Lavy (1999)



Today's lecture

The big question: How can we quantify and test the relationship between two numeric variables?

1. Motivation & data
2. The Simple Linear Regression model
3. Ordinary Least Squares estimation
- 4. Goodness of fit (R^2)**
5. Inference: standard errors & the t -test
6. Reading regression output

Goodness of Fit (R^2)

How well does the model fit?

Key question: How much of the variation in y does our model (the x variable) explain?

Recall: $y_i = \hat{y}_i + e_i$

This gives us a **variance decomposition**:

$$\underbrace{Var(y)}_{\text{Total variation}} = \underbrace{Var(\hat{y})}_{\text{Explained}} + \underbrace{Var(e)}_{\text{Unexplained}}$$

In terms of sums of squares:

$$\underbrace{\sum_{i=1}^n (y_i - \bar{y})^2}_{\text{Sum of Squares Total} = SST} = \underbrace{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}_{\text{Sum of Squares Explained} = SSE} + \underbrace{\sum_{i=1}^n e_i^2}_{\text{Sum of Squared Residuals} = SSR}$$

R^2 : the coefficient of determination

! Definition

$$R^2 = \frac{SSE}{SST} = 1 - \frac{SSR}{SST} \in [0, 1]$$

R^2 is the fraction of the total variation in y explained by the regression model.

```
1 summary(reg)$r.squared
```

```
[1] 0.04715649
```

Interpretation: Our class size variable explains 4.7% of the variation in average math scores across classes.

! Two important warnings

1. Low R^2 does **not** mean the regression is useless; in social sciences, $R^2 < 0.1$ is common. Causal identification matters more than fit.
2. High R^2 does **not** mean the relationship is causal.

R^2 and correlation (in simple regression)

In simple linear regression, R^2 equals the **squared correlation** between x and y :

$$R^2 = r_{xy}^2$$

```
1 cor(grades$classize, grades$avgmath)^2
```

```
[1] 0.04715649
```

```
1 summary(reg)$r.squared
```

```
[1] 0.04715649
```

This confirms: the model explains only about 4.7% of the variation. Class size alone is not a great predictor of math scores.

Today's lecture

The big question: How can we quantify and test the relationship between two numeric variables?

1. Motivation & data
2. The Simple Linear Regression model
3. Ordinary Least Squares estimation
4. Goodness of fit (R^2)
- 5. Inference: standard errors & the t -test**
6. Reading regression output

Inference: Standard Errors & the t -test

Why do we need inference?

Our OLS estimate $\hat{\beta}_1 = 0.317$ was computed from **one sample** of Israeli classrooms in 1991.

If we had drawn a **different sample** (different schools, different year), we would get a **different estimate**.

The fundamental question: Is the estimated relationship **real**? Could we have observed $\hat{\beta}_1 = 0.317$ simply by **chance**, even if there were truly **no** relationship between class size and math scores in the population?

To answer this, we need to know the **sampling distribution** of $\hat{\beta}_1$: how much it varies across samples.

This is exactly the framework from Lectures 4–5! The tool is the **standard error**.

The population model

We distinguish between:

	Sample	Population
Model	$y_i = \hat{\beta}_0 + \hat{\beta}_1 x_i + e_i$	$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$
Parameters	$\hat{\beta}_0, \hat{\beta}_1$ (estimated)	β_0, β_1 (unknown, true)
Errors	e_i (residuals)	ε_i (true error term)

Goal: Use $\hat{\beta}_1$ to make inference about the true population parameter β_1 .

For $\hat{\beta}_1$ to be a useful estimator of β_1 , we want two things:

Property	Meaning
Unbiasedness	$\hat{\beta}_1$ is correct on average: $E[\hat{\beta}_1] = \beta_1$
Precision	$\hat{\beta}_1$ does not vary too much across samples: small $Var(\hat{\beta}_1)$

The standard error addresses precision. But what about bias?

The variance of $\hat{\beta}_1$

One can show that:

$$Var(\hat{\beta}_1) = \frac{\sigma^2}{SST_x} = \frac{\sigma^2}{\sum_i (x_i - \bar{x})^2}$$

where $\sigma^2 = Var(\varepsilon_i)$ is the variance of the population error and $SST_x = \sum_i (x_i - \bar{x})^2$ is the total variation in x .

Intuition: $Var(\hat{\beta}_1)$ is smaller when the errors are less noisy (small σ^2) and when x varies a lot across observations (large SST_x): both make the slope easier to pin down.

Note

This formula assumes observations are **independent** and that error variance is **constant** across values of x . Violations of either are covered in econometrics. See [the formal derivation here](#) if you are curious.

Estimating σ^2 : the residual standard error

The true $\sigma^2 = \text{Var}(\varepsilon_i)$ is unknown, so we estimate it from the residuals e_i .

Why not just use $\frac{1}{n} \sum e_i^2$? Because $e_i = y_i - \hat{\beta}_0 + \hat{\beta}_1 x_i$ and we “lose” 2 degrees of freedom by estimate $\hat{\beta}_0$ and $\hat{\beta}_1$.

$$\hat{\sigma}^2 = \frac{SSR}{n - 2} = \frac{\sum_{i=1}^n e_i^2}{n - 2}$$

$$\hat{\sigma} = \sqrt{\frac{SSR}{n - 2}} \quad (\text{residual standard error})$$

Dividing by $n - 2$ instead of n makes $\hat{\sigma}^2$ an **unbiased** estimator of σ^2 .

The standard error formula

Plugging $\hat{\sigma}^2$ into the variance formula:

$$\widehat{Var}(\hat{\beta}_1) = \frac{\hat{\sigma}^2}{SST_x} = \frac{SSR/(n-2)}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

⚠ Standard error of $\hat{\beta}_1$

$$SE(\hat{\beta}_1) = \sqrt{\widehat{Var}(\hat{\beta}_1)} = \frac{\hat{\sigma}}{\sqrt{SST_x}} = \sqrt{\frac{SSR/(n-2)}{\sum_{i=1}^n (x_i - \bar{x})^2}}$$

We write $SE(\hat{\beta}_1)$ rather than $SD(\hat{\beta}_1)$ to remind ourselves this is an **estimate** of the standard deviation of $\hat{\beta}_1$'s sampling distribution.

Three drivers of $SE(\hat{\beta}_1)$

$$SE(\hat{\beta}_1) = \frac{\hat{\sigma}}{\sqrt{SST_x}} = \frac{\sqrt{SSR/(n-2)}}{\sqrt{\sum_i (x_i - \bar{x})^2}}$$

Driver	Intuition	Direction
More noise ($\uparrow \hat{\sigma}$)	Harder to isolate signal from noise	$SE \uparrow$
More spread in x ($\uparrow SST_x$)	More variation in x to pin down the slope	$SE \downarrow$
More data ($\uparrow n$)	$\hat{\sigma}$ stays stable (SSR grows, but divided by $n - 2$); $\sqrt{SST_x}$ grows like $\sqrt{n}$	$SE \downarrow$

Computing $SE(\hat{\beta}_1)$ in R

Let's verify the formula step by step:

```
1 reg    <- lm(avgmath ~ classsize, data = grades)
2 n      <- nrow(grades)
3 SSR    <- sum(residuals(reg)^2)
4 SST_x  <- sum((grades$classsize - mean(grades$classsize))^2)
5
6 sigma_hat <- sqrt(SSR / (n - 2))      # residual standard error
7 SE_beta1 <- sigma_hat / sqrt(SST_x)  # SE of slope
```

sigma_hat: 9.3702

SE(beta1) by hand: 0.0318

SE(beta1) from lm(): 0.0318

They match! The formula and `lm()` give the same answer.

Your turn! #3

10:00

Using `reg_reading <- lm(avgverb ~ classsize, grades)`:

1. Extract the vector of residuals with `residuals(reg_reading)`. Compute the $SSR = \sum_i e_i^2$.
2. Compute $\hat{\sigma} = \sqrt{SSR/(n - 2)}$ where $n = \text{nrow(grades)}$.
3. Compute $SST_x = \sum_i (x_i - \bar{x})^2$ using `classsize`.
4. Compute $SE(\hat{\beta}_1)$ using the formula. Then compare with `summary(reg_reading)$coefficients[2, 2]`.
5. **Thinking question:** Suppose you repeated the regression but restricted to only classes of size 20–25 (much less variation in x). Would you expect $SE(\hat{\beta}_1)$ to be larger or smaller? Why?

Hypothesis Testing & the t -statistic

The default hypothesis test

By default, statistical software tests:

$$H_0 : \beta_1 = 0 \quad \text{vs} \quad H_A : \beta_1 \neq 0$$

Interpretation: Under H_0 , class size has **no linear relationship** with math scores in the population.

We ask: *how likely is it to observe our estimate $\hat{\beta}_1 = 0.317$ if the true β_1 were zero?*

To answer this, we need the **sampling distribution** of $\hat{\beta}_1$ under H_0 .

The t -statistic

The general formula standardises the distance between our estimate and the hypothesised value: $t = \frac{\hat{\beta}_1 - \beta_1^0}{SE(\hat{\beta}_1)}$, where β_1^0 is the null hypothesis value.

For $H_0 : \beta_1 = 0$, this simplifies to: $t = \frac{\hat{\beta}_1}{SE(\hat{\beta}_1)}$

Interpretation: The t -statistic counts *how many standard errors* our estimate lies from zero.

Under H_0 , this follows a t -distribution with $n - 2$ degrees of freedom:

$$t = \frac{\hat{\beta}_1}{SE(\hat{\beta}_1)} \sim t_{n-2} \quad \text{under } H_0$$

Why t_{n-2} and not the standard normal?

Recall from Lecture 5: when we **estimate** σ from the data (using $\hat{\sigma}$), there is extra uncertainty.

- If σ were **known**, we'd use $\frac{\hat{\beta}_1 - \beta_1^0}{\sigma / \sqrt{SST_x}} \sim N(0, 1)$
- Since σ is **unknown** and estimated by $\hat{\sigma}$, we use a t_{n-2} distribution

Good news for large samples

When n is large (say $n > 100$), $t_{n-2} \approx N(0, 1)$. In our data with $n \approx 2,000$, the difference is negligible.

The **degrees of freedom** $n - 2$ reflect that we estimated $\hat{\beta}_0$ and $\hat{\beta}_1$ from the data (2 parameters, 2 df lost).

Computing the t -statistic

```
1 reg <- lm(avgmath ~ classsize, grades)
2 beta1_hat <- coef(reg)[2]
3 SE_beta1 <- summary(reg)$coefficients[2, 2]
4 t_stat <- beta1_hat / SE_beta1
```

beta1_hat: 0.3175

SE(beta1): 0.0318

t-stat: 9.991

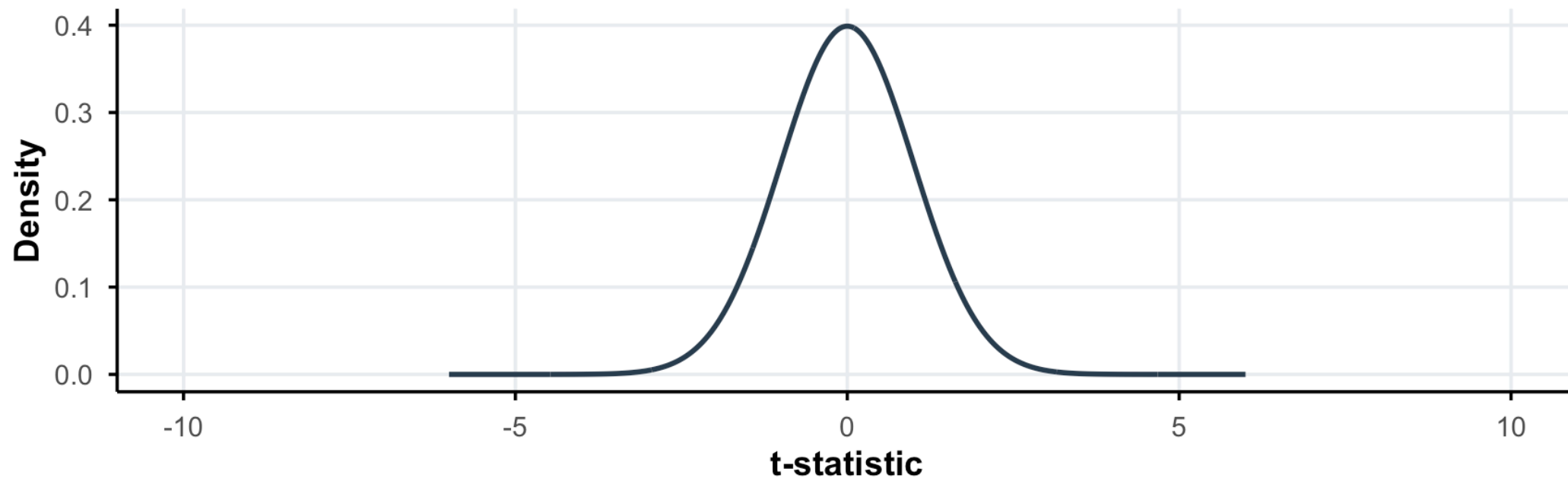
$$t = 0.3175 / 0.0318 = 9.99$$

Our estimate is **10 standard errors** away from zero.

The p -value

Sampling distribution of t under $H_0: \beta_1 = 0$

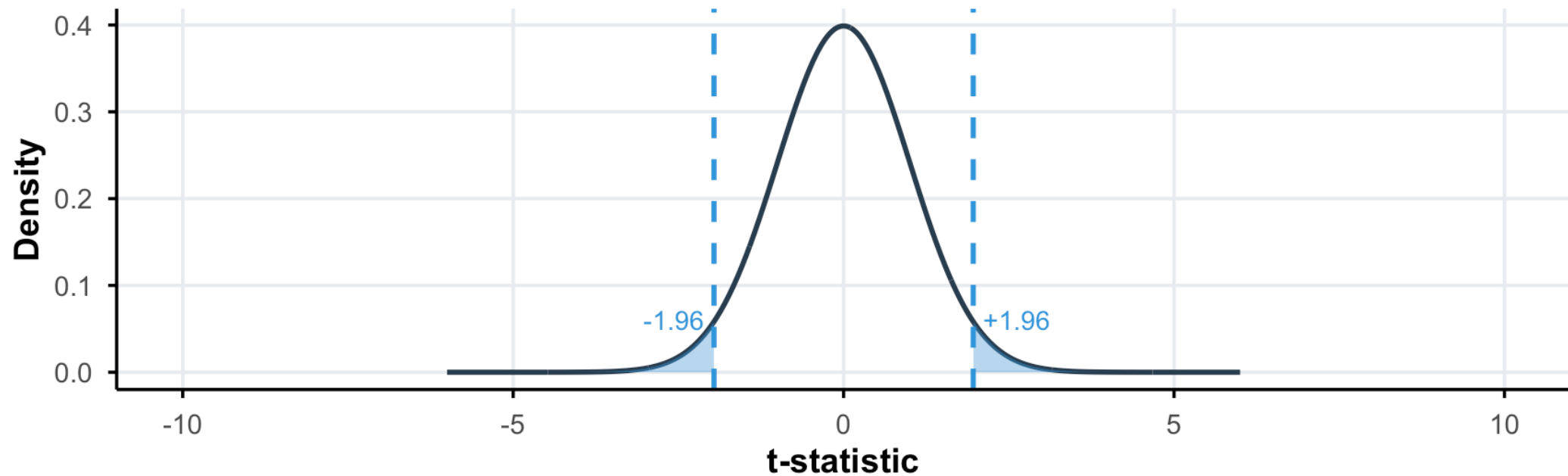
If H_0 is true, our t -statistic is just one draw from this distribution



The p -value

Rejection region at $\alpha = 5\%$

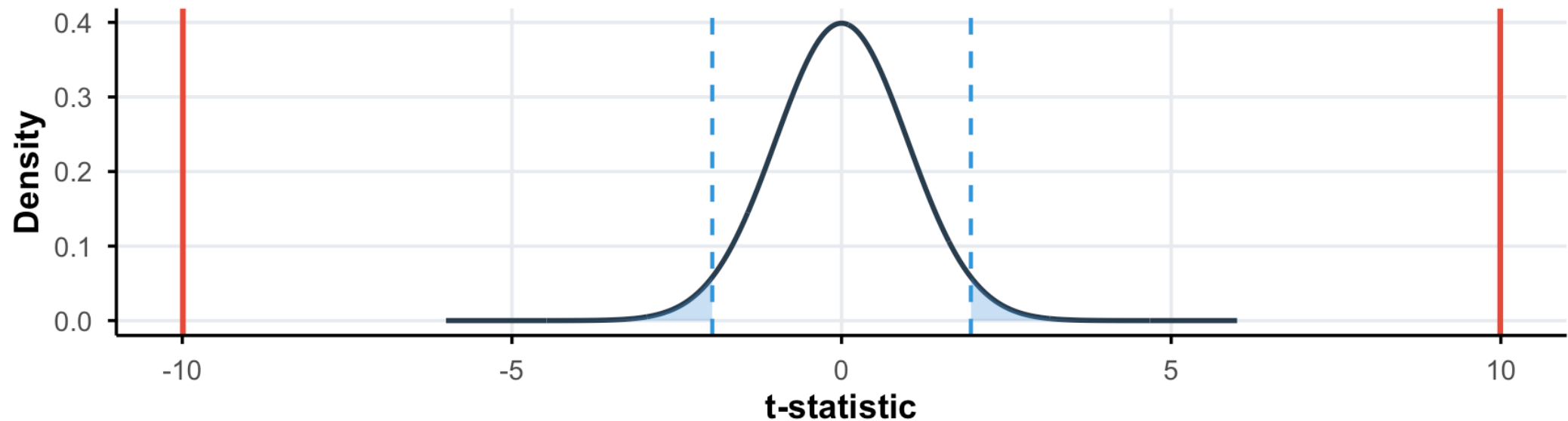
Blue shaded area = 5% — we reject H_0 if $|t|$ falls in this region



The p -value

Our observed t-statistic vs. the rejection region

Red area = p-value | Blue dashes = ± 1.96 (5% critical values)



$$p\text{-value} = P(|t_{n-2}| \geq |t_{\text{obs}}|)$$

```
1 p_value <- 2 * pt(-abs(t_stat), df = df)
```

p-value: 5.65e-23

Decision rule and conclusion

At significance level $\alpha = 0.05$:

p-value	5.6e-23
Reject H_0 ?	Yes – $p < 0.05$
Conclusion	There is strong statistical evidence that class size is significantly associated with average math scores

Rule of thumb: If $|t| > 2$ (more precisely, $|t| > t_{0.025, n-2}^* \approx 1.96$ for large n), we reject H_0 at the 5% level.

Here $|t| = 10$: well above 2.

Warning

Statistical significance $\neq$ practical significance. A coefficient can be statistically significant but economically tiny (especially with large n). Always report the magnitude and the SE alongside the p-value!

Confidence interval for β_1

A 95% CI for β_1 :

$$CI_{95\%} = \left[\hat{\beta}_1 \pm t_{0.025, n-2}^* \cdot SE(\hat{\beta}_1) \right]$$

For large n , $t_{0.025, n-2}^* \approx 1.96$.

```
1 t_crit <- qt(0.975, df = n - 2)
2
3 CI_low  <- beta1_hat - t_crit * SE_beta1
4 CI_high <- beta1_hat + t_crit * SE_beta1
```

95% CI: [0.255 , 0.38]

Interpretation: We are 95% confident that the true β_1 lies between 0.255 and 0.38.

Notice: the CI does **not** include 0, which is consistent with rejecting $H_0 : \beta_1 = 0$ at the 5% level.

Your turn! #4

12:00

Now apply everything we covered to the **verbal (reading) score** regression.

1. Run `reg_verb <- lm(avgverb ~ classsize, data = grades)` and print `summary(reg_verb)`.
2. From the output, identify the estimate, SE, t -statistic, and p-value for `classsize`. What do they tell you?
3. Verify the t -statistic manually: compute `coef(reg_verb)[2] / summary(reg_verb)$coefficients[2, 2]`.
4. Compute a **95% confidence interval** for β_1 by hand using `qt(0.975, df = nrow(grades) - 2)`.
5. Is the effect of class size on verbal scores statistically significant at the 5% level? How does it compare in magnitude to the math regression?
6. (*Bonus*) A classmate argues: “Since both coefficients are positive, larger classes **cause** better outcomes.” What is wrong with this interpretation? Name at least one potential confounder.

Today's lecture

The big question: How can we quantify and test the relationship between two numeric variables?

1. Motivation & data
2. The Simple Linear Regression model
3. Ordinary Least Squares estimation
4. Goodness of fit (R^2)
5. Inference: standard errors & the t -test
6. Reading regression output

Reading Regression Output

summary() output

```
1 summary(reg)
```

Call:

```
lm(formula = avgmath ~ classsize, data = grades)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-38.359	-6.039	0.356	6.593	27.564

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	57.79392	0.97365	59.358	<2e-16 ***
classsize	0.31749	0.03178	9.991	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 9.37 on 2017 degrees of freedom

Multiple R-squared: 0.04716, Adjusted R-squared: 0.04668

F-statistic: 99.82 on 1 and 2017 DF, p-value: < 2.2e-16

Anatomy of the regression table

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	57.7939	0.9736	59.3581	0
classize	0.3175	0.0318	9.9911	0

Column	Meaning
Estimate	OLS coefficient $\hat{\beta}_k$
Std. Error	$SE(\hat{\beta}_k)$: estimated SD of the sampling distribution of $\hat{\beta}_k$
t value	$t = \hat{\beta}_k / SE(\hat{\beta}_k)$: test statistic for $H_0 : \beta_k = 0$
Pr(> t)	Two-sided p-value for $H_0 : \beta_k = 0$

Significance codes: *** ($p < 0.001$), ** ($p < 0.01$), * ($p < 0.05$), . ($p < 0.1$)

`tidy()` from `broom`: cleaner output

```
1 library(broom)
2 tidy(reg, conf.int = TRUE, conf.level = 0.95)
```

```
# A tibble: 2 × 7
  term          estimate std.error statistic  p.value conf.low conf.high
<chr>         <dbl>     <dbl>     <dbl>    <dbl>   <dbl>   <dbl>
1 (Intercept)   57.8       0.974      59.4    0        55.9    59.7
2 classsize     0.317     0.0318     9.99 5.65e-23    0.255    0.380
```

`tidy()` returns a data frame: much easier to work with programmatically.

The `conf.low` and `conf.high` columns give the 95% CI directly.

Putting it all together: Angrist & Lavy

```
1 reg_math <- lm(avgmath ~ classsize, grades)
2 reg_verb <- lm(avgverb ~ classsize, grades)
3
4 tidy(reg_math, conf.int = TRUE) %>% filter(term == "classsize")
```

```
# A tibble: 1 × 7
  term      estimate std.error statistic  p.value conf.low conf.high
<chr>      <dbl>     <dbl>     <dbl>   <dbl>   <dbl>   <dbl>
1 classsize  0.317      0.0318      9.99 5.65e-23  0.255   0.380
```

```
1 tidy(reg_verb, conf.int = TRUE) %>% filter(term == "classsize")
```

```
# A tibble: 1 × 7
  term      estimate std.error statistic  p.value conf.low conf.high
<chr>      <dbl>     <dbl>     <dbl>   <dbl>   <dbl>   <dbl>
1 classsize  0.222      0.0256      8.69 7.51e-18  0.172   0.272
```

Math: $\hat{\beta}_1 = 0.317$ (SE = 0.032, $p < 0.001$)

Verbal: $\hat{\beta}_1 = 0.222$ (SE = 0.026)

Both positive and statistically significant; but this doesn't mean larger classes *cause* better outcomes (confounders!).

Wrapping up

What we covered today

Simple Linear Regression: the full workflow

1. **Model:** $y_i = \hat{\beta}_0 + \hat{\beta}_1 x_i + e_i$: summarises a linear relationship with a line
2. **OLS:** Minimises $\sum e_i^2 \rightarrow$ unique formulas for $\hat{\beta}_0$ and $\hat{\beta}_1$
3. **Goodness of fit:** $R^2 = 1 - SSR/SST$: fraction of variation explained
4. **Standard error:** $SE(\hat{\beta}_1) = \hat{\sigma} / \sqrt{SST_x}$: quantifies sampling uncertainty
5. **t-test:** $t = \hat{\beta}_1 / SE(\hat{\beta}_1) \sim t_{n-2}$ under $H_0 : \beta_1 = 0$
6. **Confidence interval:** $\hat{\beta}_1 \pm t_{0.025}^* \cdot SE(\hat{\beta}_1)$

Looking ahead

Today's limitation: we only included class size. But other variables (% disadvantaged students, school type, etc.) are correlated with both class size *and* performance.

→ **Next year: Multiple regression:** control for multiple variables simultaneously, isolate the effect of class size, and understand **omitted variable bias** formally.

The bigger picture

Regression is a tool for *description* and *prediction*. To make **causal claims**, we need careful research designs: randomised experiments, natural experiments, instrumental variables, regression discontinuity... This is the realm of **econometrics** and **causal inference**!

Thanks for a great semester!